

[YIJC/2025/Q4]

- 4 A programmer implemented a queue data structure with Object-Oriented Programming (OOP) to design a messaging system, Message of the Day (MOTD), for teachers to create and broadcast announcements that will be shown to students on their personal learning devices (PLDs).

The class `Message` contains three attributes:

- `title` the title of the message
- `content` the contents of the message
- `next` points to the next `Message` object.

Task 4.1

Write program code to declare the class `Message`, its constructor, and its methods. [3]

The class `MOTD` contains one attribute:

- `root` the pointer pointing to the first message object in the queue.

The class `MOTD` has the following methods initially:

- a constructor that initialises the pointer `root` to `None` when the queue is empty.
- `add_msg(title, content)` adds a `Message` object to the queue and returns `True`
- `display()` prints an ordered list of the titles of the messages, according to the `Message` objects in the queue. If there are no messages in the queue, print 'No message in system.'

Task 4.2

Write program code to declare the:

- class `MOTD`
- constructor, `add_msg()`, and `display()` methods. [7]

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The table below contains the messages to be added to the MOTD queue according to the following order:

	Title	Content
1	'Title A'	'Content A'
2	'Title B'	'Content B'
3	'Title C'	'Content C'
4	'Title D'	'Content D'
5	'Title E'	'Content E'

Task 4.3

Write program code to:

- initialise a new MOTD queue as `motd1`
- add the five messages to the `motd1` queue

Test your program and use the `display()` method to display all the messages in the `motd1` queue.

Test Case

```
>>> motd1.display()
```

Output:

Title A -> Title B -> Title C -> Title D -> Title E [2]

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To ensure that more important messages are displayed first, a priority rating ranging from 1 to 5 is added in the MOTD system. The value 1 denotes the most important and 5 denotes the least important. If there are two or more messages with the same priority rating, the messages will be displayed according to the order they were added to the MOTD queue.

When the students log in to the computer network, the MOTD application will display the messages automatically on their PLDs.

Task 4.4

Modify the class `Message` by adding a new attribute `priority` which is the priority rating of the message, and its associated methods.

Modify the class `MOTD` by adding an additional method:

- `add_msg_priority(title, content, priority)` adds a `Message` object to the queue in the order of the priority, from 1 to 5, and returns `True`
- `broadcast()` removes the messages one by one according to their order in the queue and display the messages, with the titles and contents. [9]

The table below contains the messages to be added to the MOTD queue according to the following order:

	Title	Content	Priority
1	'Title A'	'Content A'	5
2	'Title B'	'Content B'	3
3	'Title C'	'Content C'	1
4	'Title D'	'Content D'	2
5	'Title E'	'Content E'	3

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Task 4.5

Write program code to:

- initialise a new MOTD queue as `motd2`
- use `add_msg_priority()` to add the five messages to the `motd2` queue

Test your program by:

- calling `display()` on `motd2` queue
- calling `broadcast()` on `motd2` queue
- calling again `display()` on `motd2` queue

The following three test cases are to be executed consecutively:

Test Case

```
>>> motd2.display()
```

Output:

```
Title C -> Title D -> Title B -> Title E -> Title A
```

Test Case

```
>>> motd2.broadcast()
```

Output:

```
1. Title C: Content C
2. Title D: Content D
3. Title B: Content B
4. Title E: Content E
5. Title A: Content A
Last message printed.
```

Test Case

```
>>> motd2.display()
```

Output:

```
No messages in system.
```